

Exp 30-Group-wise Score Comparison of Men and Women Using a Bar Plot

Aim

To write a Python program using Matplotlib to create a grouped bar plot showing the scores of men and women for different groups (G1–G5) using multiple X-values on the same chart.

Procedure

1. Import the required `matplotlib.pyplot` library.
2. Define the scores of men and women.
3. Define the X-axis positions for the groups G1 to G5.
4. Set the width of each bar.
5. Plot the men's scores using one set of X-values.
6. Plot the women's scores beside the men's bars using adjusted X-values.
7. Add X-axis label, Y-axis label, title, legend, and grid.
8. Display the bar chart using `plt.show()`.

Code

```
30.py - C:/Users/subha/OneDrive/Desktop/Query processing/30.py (3.12.10)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
import numpy as np

# Sample data
means_men = (22, 30, 35, 35, 26)
means_women = (25, 32, 30, 35, 29)

# X-axis positions
groups = ['G1', 'G2', 'G3', 'G4', 'G5']
x = np.arange(len(groups))

# Width of bars
width = 0.35

# Create bar plot
plt.bar(x - width/2, means_men, width, label='Men', color='green')
plt.bar(x + width/2, means_women, width, label='Women', color='red')

# Labels and title
plt.xlabel('Group')
plt.ylabel('Scores')
plt.title('Scores by Group and Gender')

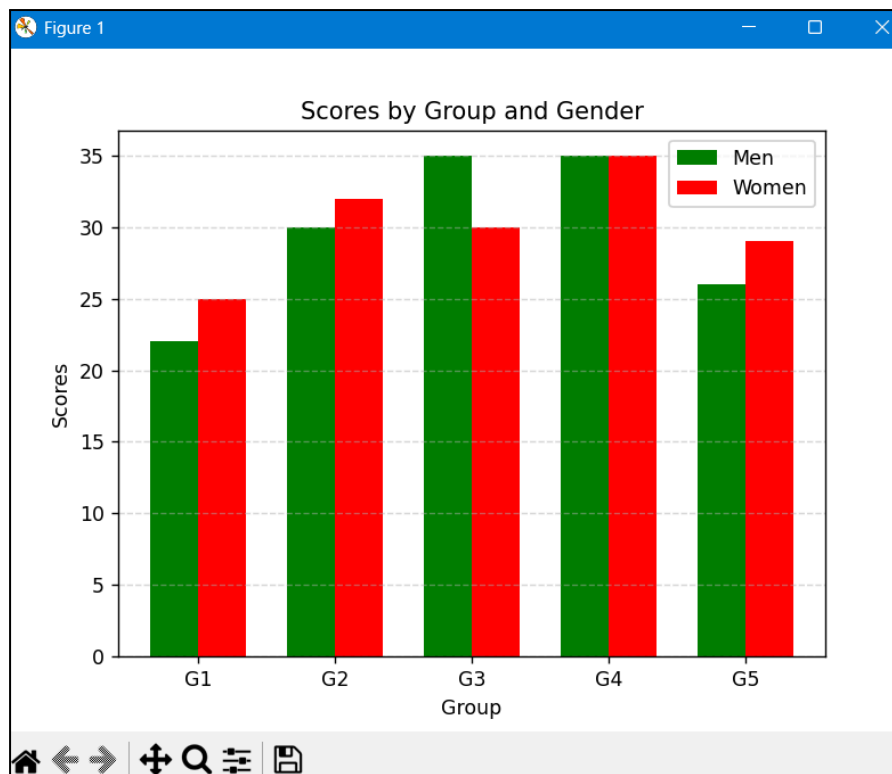
# X-axis labels
plt.xticks(x, groups)

# Legend
plt.legend()

# Grid
plt.grid(axis='y', linestyle='--', alpha=0.5)

# Display plot
plt.show()
```

Output



Result

Thus, the Python program successfully creates a grouped bar plot comparing the scores of men and women across five groups using Matplotlib.

31. Stacked Bar Plot of Group-wise Scores with Error Bars

Aim

To write a Python program using Matplotlib to create a stacked bar plot with error bars, where the women's bars are stacked on top of the men's bars.

Procedure

1. Import `matplotlib.pyplot` and `numpy`.
2. Define the mean scores of men and women.
3. Define the standard deviations for men and women.
4. Create X-axis positions for the five groups.
5. Plot the men's scores with their standard deviation as error bars.
6. Use the men's scores as the bottom value for the women's bars.
7. Plot the women's scores with their standard deviation as error bars.
8. Add title, axis labels, legend, and grid.
9. Display the stacked bar chart.

Code

```
31.py - C:/Users/subha/OneDrive/Desktop/Query processing/31.py (3.12.10)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
import numpy as np

# Sample data
means_men = (22, 30, 35, 35, 26)
means_women = (25, 32, 30, 35, 29)

men_std = (4, 3, 4, 1, 5)
women_std = (3, 5, 2, 3, 3)

# Groups
groups = ['Group1', 'Group2', 'Group3', 'Group4', 'Group5']

# X-axis positions
x = np.arange(len(groups))

# Create stacked bar plot
plt.bar(x, means_men,
        yerr=men_std,
        capsize=4,
        color='red',
        label='Men')

plt.bar(x, means_women,
        bottom=means_men,
        yerr=women_std,
        capsize=4,
        color='green',
        label='Women')

# Labels and title
plt.xlabel('Groups')
plt.ylabel('Scores')
plt.title('Scores by group and gender')

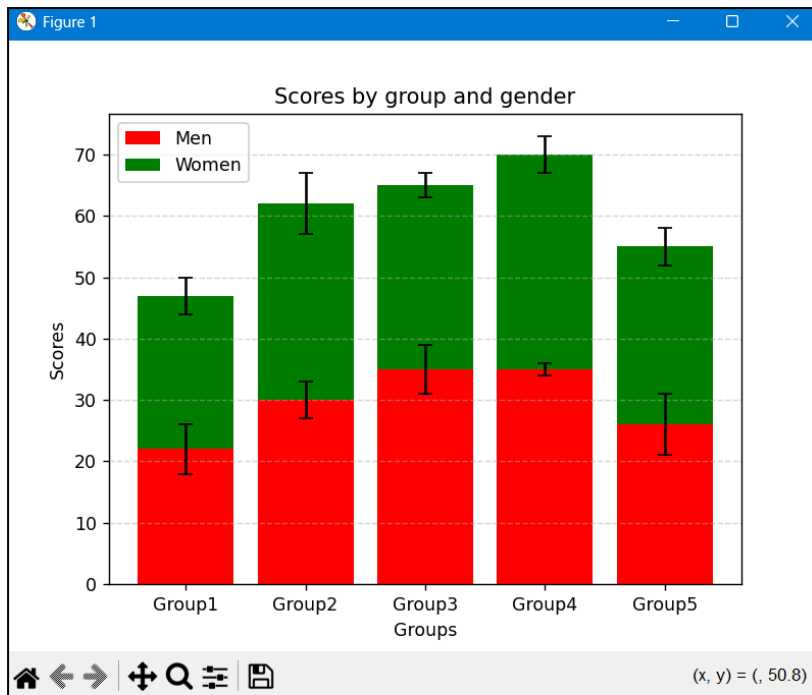
# X-axis labels
plt.xticks(x, groups)

# Legend
plt.legend()

# Grid
plt.grid(axis='y', linestyle='--', alpha=0.5)

# Display graph
plt.show()
```

Output



Result

Thus, the Python program successfully creates a stacked bar plot with error bars, with women's scores stacked on top of men's scores using Matplotlib.

Aim

To write a Python program using Matplotlib to generate a scatter plot by taking random distributions of X and Y and plotting them against each other.

Procedure

1. Import the required numpy and matplotlib.pyplot libraries.
2. Generate random values for X and Y using np.random.randn().
3. Create a scatter plot using plt.scatter().
4. Set the X-axis and Y-axis labels.
5. Add a suitable title to the graph.
6. Display the scatter plot using plt.show().

Code

```
32.py - C:/Users/subha/OneDrive/Desktop/Query processing/32.py (3.12.10)
File Edit Format Run Options Window Help
import numpy as np
import matplotlib.pyplot as plt

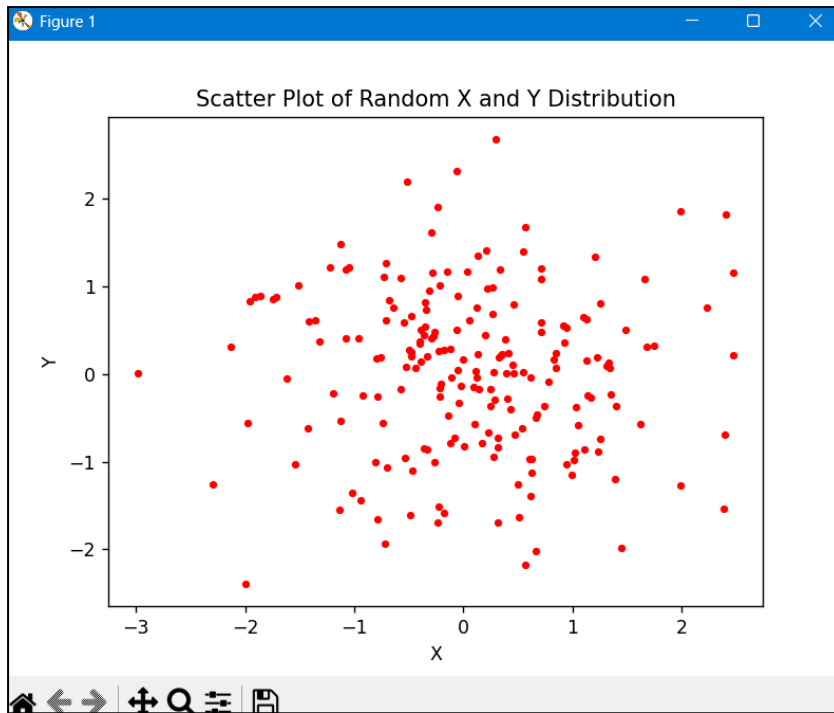
# Generate random data
np.random.seed(10)
x = np.random.randn(200)
y = np.random.randn(200)

# Create scatter plot
plt.scatter(x, y, color='red', s=10)

# Add labels and title
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot of Random X and Y Distribution')

# Display plot
plt.show()
```

Output



Result

Thus, the Python program successfully generates a scatter plot of randomly distributed X and Y values using Matplotlib.